

ARTICLE

Measuring biological effectiveness across a very large, coherent network of coastal marine protected areas

Barbara Spiecker¹  | Mark H. Carr²  | Daniel P. Malone²  |
Avrey Parsons-Field¹ | Kathryn Koehn¹ | Daniel J. Pondella II³ |
Jennifer E. Caselle¹

¹Marine Science Institute, University of California Santa Barbara, Santa Barbara, California, USA

²Department of Ecology and Evolutionary Biology, University of California Santa Cruz, Santa Cruz, California, USA

³Vantuna Research Group, Occidental College, Los Angeles, California, USA

Correspondence

Barbara Spiecker

Email: barbara.spiecker@gmail.com

Funding information

David and Lucile Packard Foundation;
Gordon and Betty Moore Foundation;
California Sea Grant, Grant/Award
Numbers: R/MPA-43, R/MPA-49A;
California Ocean Protection Council

Handling Editor: José M. Riascos

Abstract

We evaluated the performance of California's coastal marine protected area (MPA) network, the largest scientifically designed network of its kind, using SCUBA survey data from a large-scale, long-term kelp forest monitoring program. To comprehensively evaluate MPA performance, we employed multiple metrics across various scales of taxonomic and spatial aggregation and four key results emerged. First, population responses to MPAs varied greatly across the large network, whether evaluated as combined taxa or individual focal species or at single MPA or regional scales. Second, while responses varied widely, effect size analysis revealed differences in magnitude of MPA performance over time, with targeted species increasing or declining less within MPAs compared to non-targeted species in all regions except the North Coast. Third, average response ratios further supported these findings by providing a cumulative perspective on MPA performance. Specifically, 80% of MPAs showed positive response ratios for targeted species, with 12 of these cases being significant, indicating generally higher biomass inside MPAs. Finally, using our most conservative performance framework of divergence between MPA and non-MPA (Ref) sites and between targeted and non-targeted species, only the South Coast and Northern Channel Islands regions exhibited a significantly positive trend in biomass ratios (MPA/Ref) over time for targeted species. While biomass ratios increased significantly over time within MPAs for targeted species, divergence between targeted and non-targeted species' trends was largely absent, except in the Northern Channel Islands where a marginally non-significant effect was noted. Taken together,

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Ecological Applications* published by Wiley Periodicals LLC on behalf of The Ecological Society of America.

these results demonstrate that MPA performance is variable across regions and taxa, and different aspects of performance were revealed depending on the metric used. This underscores the need for comprehensive evaluation approaches when assessing the effectiveness of large, networked MPAs.

KEYWORDS

algae, California, fish, invertebrate, kelp forest, marine protected area, marine reserve, MPA network, regional management

INTRODUCTION

Global interest in and use of marine protected areas (MPAs) for the purpose of biodiversity conservation and fisheries management has increased greatly over the past two decades as international governing bodies set growing global targets for their spatial coverage (Boonzaier & Pauly, 2016; Lubchenco & Grorud-Colvert, 2015). These include the Convention on Biological Diversity (CBD)'s Aichi Target 11 to protect 10% of the ocean by 2020, the United Nations Sustainable Development Goal 14.5, and the recent adoption of CBD's Kunming-Montreal Global Biodiversity Framework (GBF) to set aside at least 30% of the ocean in a network of highly protected MPAs by 2030 (30×30) (Jones et al., 2020; O'Leary et al., 2016). Meeting the conservation and fisheries objectives intended of MPAs requires embedding protected areas of appropriate levels of protection (Grorud-Colvert et al., 2021) within coastal areas that often experience intensive and diverse economically and culturally significant uses.

A growing literature suggests that ecological networks of MPAs are the most promising means of simultaneously achieving conservation goals while accommodating or enhancing diverse ocean uses (e.g., IUCN World Commission on Protected Areas [IUCN-WCPA], 2008). By themselves, small MPAs may not support biodiversity protection goals or populations that are large enough to sustain themselves or meet the area-based goals of Aichi 11 or 30×30 . However, in some regions, social and economic constraints can make it impractical and/or inequitable to create single MPAs that are large enough to achieve goals. Establishing well-designed networks of smaller MPAs connected by the transfer of propagules (i.e., spores, eggs, larvae), juveniles, or adults may reduce economic impacts without compromising conservation benefits (Boero, 2017; Carr et al., 2017; Gaines et al., 2010; Olsen et al., 2013). Networks of MPAs potentially provide even more benefits than single MPAs by encompassing greater habitat diversity and redundancy for multiple species and life-stages (McCook et al., 2010). For example, a network of MPAs can include deeper offshore habitats required of adults for feeding and reproduction and inshore nursery grounds required

of juveniles. If the network extends over a broad geographical area and along a gradient of climatic regimes, it can provide more resilience to climate stressors than would a concentration of MPAs in one or a few places (Carr & Hazen, 2019; McLeod et al., 2009; Mumby et al., 2011; Roberts et al., 2017; Steneck et al., 2009). Thus, networks can offer a better compromise between human use and conservation than single large, protected areas.

Coherent, scientifically designed networks of MPAs are starting to be developed around the world, but with few exceptions, little empirical evidence for benefits across networks exists (Baetscher et al., 2019; Ballantine, 2014; Batista & Cabral, 2016; Fenberg et al., 2012; Hamilton et al., 2010; Harrison et al., 2012; Jessen et al., 2017; Knott et al., 2021; Lowry et al., 2009; Smith et al., 2025; Starr et al., 2015; Ziegler et al., 2023). One reason for the paucity of empirical MPA network assessments is simply due to the logistic and financial challenges of monitoring multiple ecosystems over large spatial scales, often encompassing several bioregions or biogeographic breaks (Addison et al., 2017; Ahmadi et al., 2015; Gill et al., 2017; Knott et al., 2021; Smith et al., 2025). Spatial variation in species abundances and composition, reflecting changes in environmental conditions across bioregions, causes inherent challenges for evaluating ecological responses, especially for analyses that involve replication at the scale of MPAs and associated reference areas (Hamilton et al., 2010). However, evaluating MPA performance across a network encompassing diverse environmental conditions and species assemblages generates valuable insights into factors that influence ecological responses and MPA performance.

California has one of the largest, scientifically designed ecological networks of coastal MPAs in the world, with approximately 16% of its 1350-km coastline under some form of protection (Figure 1). The network of 124 MPAs in California currently accounts for approximately 57% of the total number of no-take MPAs within the continental United States ("The MPA Inventory," 2020; Van Diggelen et al., 2022). While the network encompasses all major habitats, kelp forests and rocky reefs, the focus of this study by design, were particularly important—in part

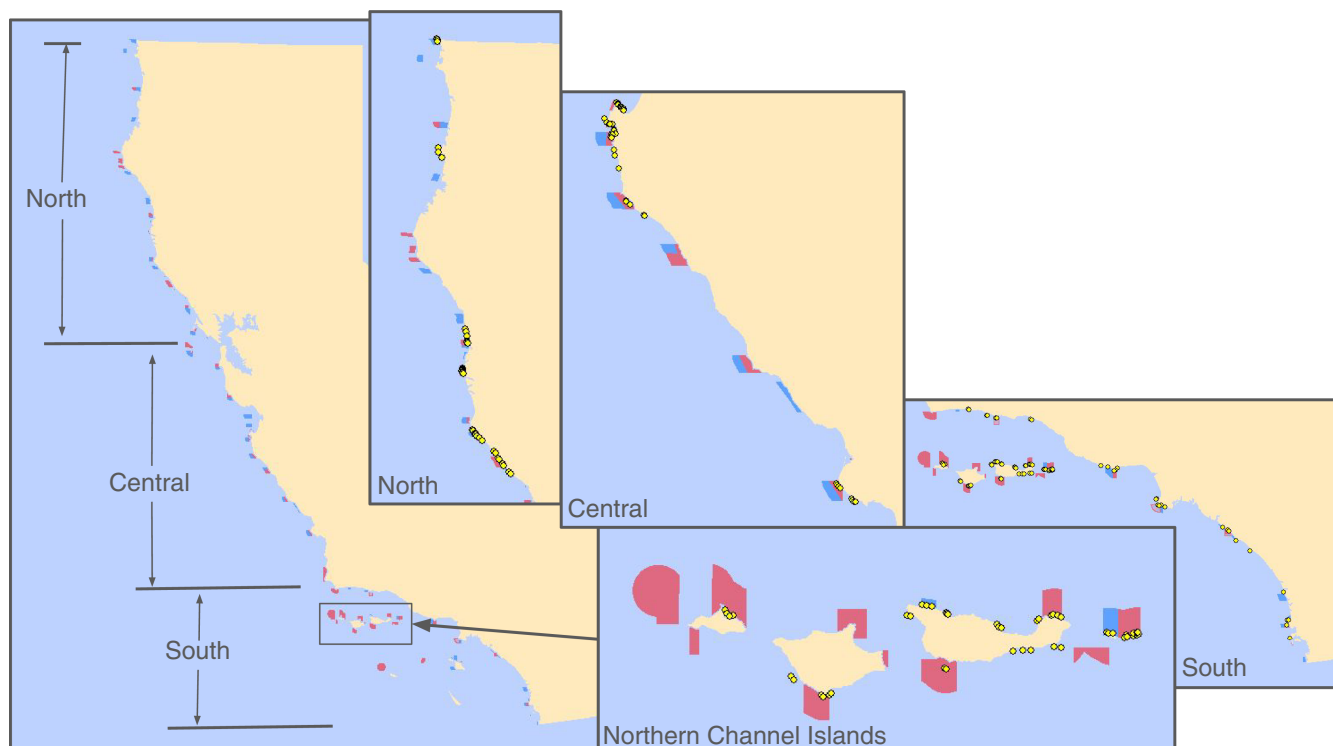


FIGURE 1 Map of the California coast with inset maps showing the divisions between monitoring regions (North, Central, and South Coasts and Northern Channel Islands). Marine protected areas are indicated in each panel: red areas generally represent State Marine Reserves, which ban all forms of take, and blue areas represent State Marine Conservation Areas, which permit some forms of take. Long-term monitoring sites used in this study are indicated in each regional inset map by yellow dots.

because they are one of the most well-studied marine systems in California and in part because of their ecological and economic importance to the state. California kelp forest and nearshore rocky reef ecosystems are among the most productive and species rich in the world and provide numerous important services in the form of recreational and commercial fisheries, non-consumptive recreational opportunities, cultural significance, and maintenance of biodiversity (Carr & Reed, 2016; Schiel & Foster, 2015). Across the globe (e.g., Krumhansl et al., 2016; Wernberg et al., 2019) and the west coast of North America (Beas-Luna et al., 2020; McPherson et al., 2021; Smith et al., 2024), kelp forests have become increasingly threatened by multiple stressors that are exacerbated by climate change, and MPAs have been suggested as a potential conservation strategy (Magris et al., 2014; McLeod et al., 2009).

We leverage a large-scale, long-term kelp forest monitoring program in California to comprehensively evaluate the performance of an MPA network in achieving its conservation objectives across an expansive (1350 km) geographic range. Specifically, we apply multiple metrics across various scales of taxonomic and spatial aggregation to assess population trajectories of both aggregated (i.e., targeted vs. non-targeted) and individual (species-specific) populations of kelp forest fishes, invertebrates,

and macroalgae in response to the establishment of MPAs across the network.

METHODS

California marine ecoregions

Kelp forest ecosystems are distributed along the coast of California from the Oregon to the Mexican border and all offshore islands. The MPA network design considered differences in the species composition and community structure of kelp forests along the coast and a geographic pattern delineating northern (Oregon border to San Francisco Bay), central (San Francisco Bay to Point Conception), and southern (Point Conception to the Mexican border) regions (Botsford et al., 2014; Gleason et al., 2013; Saarman et al., 2013). California “ecoregions” were well characterized for some time (Allen et al., 2006; Blanchette et al., 2008) and have been confirmed by subsequent studies (Beas-Luna et al., 2020; Carr & Reed, 2016). As such, the design of the network distributed MPAs across the three ecoregions to protect the breadth of diversity and functioning of these ecosystems. These distinct ecoregions are driven by differences in coastal geomorphology

(e.g., vertical relief and rugosity, turbidity, substrate composition), oceanographic conditions (e.g., water temperature, coastal upwelling), and differences between islands and mainland coast (Beas-Luna et al., 2020; Carr & Reed, 2016). Embedded in this large-scale regional geographic variation across California, smaller scale geographic differences (i.e., Northern Channel Islands vs. southern coastal mainland) must also be accounted for in the monitoring design, analyses, and evaluation of MPA performance (Hamilton et al., 2010; Pondella et al., 2005; Zahn et al., 2016). Therefore, our analyses consider four regions: North Coast, Central Coast, South Coast, and Northern Channel Islands (Figure 1).

SCUBA surveys

Monitoring of kelp forest communities has been conducted using standardized SCUBA visual survey protocols since 1999. To characterize the ecological community at each site, we conduct two types of diver surveys: (1) density and size distribution of all conspicuous fishes are recorded along transects at three levels in the water column and (2) density of large invertebrates and stipitate algae identified to species is recorded along swath transects on the reef. More details for each of these methods are described in Hamilton et al. (2010), Caselle et al. (2015), and Malone et al. (2022).

Fish surveys

Counts and size of all conspicuous fishes are visually recorded along replicate $30\text{ m} \times 2\text{ m} \times 2\text{ m}$ transects ($n = 8\text{--}12$ transects per site). At each location, transects were conducted at multiple levels in the water column: bottom, midwater, and kelp canopy (when present). Transect locations are laid out in a stratified random design, with 3–4 non-permanent transects located along an isobath at 2–4 fixed depth strata from the outer to middle to inner edge of the reef at the target depths between 5 and 20 m. At each level in the water column, a SCUBA diver identifies and counts all fishes (excluding small cryptic fishes) while estimating fish total length to the nearest centimeter.

Length estimates of fishes were converted to weights based on allometric relationships found in published and web-based (e.g., www.fishbase.org) sources. In the cases where length–weight information did not exist for a given species, the parameters from congeners with similar body size and shape were used. For each fish species, we summed weights over the bottom and midwater transects (expressed as the 60 m^2 of reef surface) and calculated the biomass density per transect area (in kilograms per 60 m^2).

Algae and invertebrate surveys

Counts of conspicuous, individually distinguishable macroalgae and macroinvertebrates (i.e., organisms larger than 2.5 cm) are visually recorded along replicate $30\text{ m} \times 2\text{ m}$ transects ($n = 4\text{--}6$ transects per site). For select species (e.g., sea urchins), very high counts are subsampled to allow extrapolation to a 60 m^2 area. Benthic transects are laid out in a stratified random design, with multiple non-permanent transects located along an isobath at 3 fixed strata (i.e., outer, middle, and inner edges of the reef, corresponding to target depths of 5, 12.5, and 20 m). Density is expressed as the number of individuals per 60 m^2 .

Spatial sampling design

To characterize species abundances within an MPA and adjacent reference (hereafter, Ref) sites outside of each MPA, surveys were conducted at 2–4 replicate sites within and 2–4 sites adjacent to each MPA (numbers of sites are provided in Appendix S1: Table S1). We averaged the transect values (per 60 m^2) at each site and then took the mean across replicate sites to generate a single average biomass density (fishes) or density estimate (invertebrates/algae) for each MPA and Ref area.

Selection of MPAs

We visually inspected data for all sites for all years to select individual MPAs with sufficient temporal coverage to allow the analysis of temporal trends. Criteria included (1) only MPAs with paired Ref sites where both were sampled in the same year, (2) at least two years of data in a “baseline period” (at or just after MPA implementation) and 3 years of data in a “recent” period (i.e., 2018–2021), and (3) more than one replicate site within the MPA and the Ref area. With these criteria applied, we had 20 MPA/Ref pairs for analysis (Appendix S1: Table S1). These included four MPAs in the North Coast region, five in the Central Coast region, four in the South Coast region, and seven in the Northern Channel Islands region. These MPAs were composed of “no-take” State Marine Reserves (SMR) and “partial-take” State Marine Conservation Areas (SMCA).

Taxonomic groupings and focal species selection

A central focus of our analytical framework is to compare the responses of species affected by the MPAs (targeted species) to species not directly affected (non-targeted species)

(as was done by Caselle et al., 2015; Hamilton et al., 2010; Ovando et al., 2021; Smith et al., 2025; Watson et al., 2007; Woodley et al., 2010). It is assumed that non-targeted species will reflect any changes in the environment similarly inside and outside of the MPA, while targeted species will reflect changes in fishing mortality. Thus, non-targeted species may act as controls for changing environmental conditions within MPAs and between MPA and reference areas, independent of MPA effects—at least early after MPA implementation. We categorized all fish species into “targeted” or “non-targeted” groups (Appendix S1: Tables S2 and S3) using information on fishing from the literature, California Department of Fish and Wildlife reports, and other sources including local knowledge. We also selected individual focal fish, invertebrate, and algae species for analysis and presentation using criteria derived from the CA State MPA Action Plan (Marine Protected Area Monitoring Action Plan, 2018) and Decadal Evaluation Working Group report (Hall-Arber et al., 2021) including those with high abundance, ecological importance, protection status, threatened or endangered status, fisheries importance, and cultural importance (Appendix S1: Table S4). Despite some kelp harvest in California, we consider the focal algae species as non-targeted for these analyses due to the low and localized nature of current extraction.

Analyses

Many environmental, ecological, and anthropogenic factors simultaneously influence species' responses to the establishment of an MPA. To separate the effects of these factors from the effects of fishing (the MPA effect), we applied several analyses and metrics to comprehensively assess MPA performance across varying levels of taxonomic and spatial aggregation, using increasingly conservative criteria for detecting MPA effects.

Average response ratios

We calculated log response ratios ($\log_{10}(\text{MPA}/\text{Ref})$) using average biomass (fishes) or density (invertebrates and algae) over the entire time series at each MPA for targeted and non-targeted fish species and for each focal fish, invertebrate, and algae species. For each survey year, biomass or density estimates were averaged among all transects surveyed at each MPA and Ref area. After calculating the mean response ratios across years, we ran a series of one-sample *t* tests to determine whether the mean response ratio for each focal species and MPA is significantly different from zero. We also used these average response ratios to explore the influence of a set of

MPA design attributes, including MPA location, MPA size, habitat diversity and richness, amount of rocky reef habitat in the MPA, and distance to the nearest port as a proxy for fishing effort. We tested the effect of these MPA traits on the average response ratio for targeted fish biomass using linear regression.

Relative change in species responses

To evaluate change in the magnitude of the difference between MPA and Ref responses over time, we calculated the slope of biomass density over years for MPA and Ref sites within each region. We did this for both targeted and non-targeted fishes and each focal fish, invertebrate, and algae species. These slopes were then compared using Hedges' *g*, which was implemented due to unequal sample sizes (i.e., numbers of sites) between MPA and Ref. To calculate percentile gain, we translated each Hedges' *g* value to a point on the normal distribution, and the resulting proportion value was then subtracted by 0.50 (mean = the score at the 50th percentile) and multiplied by 100 (Marzano et al., 2001).

Regional fish biomass by target status

We also assessed whether MPA effects on fish biomass could be detected on a year-by-year basis at the regional scale. We calculated $\log_{10}$ biomass response ratios for each region using the observed MPA and Ref transect estimates. We then used bootstrap resampling of the transect estimates to produce a distribution of possible “no difference” log biomass response ratios (i.e., fish biomass was not different inside and outside of the MPA) and tested whether these observed ratios were significantly different from the simulated “no difference” ratios. The transect values from both MPA and Ref areas within a region were pooled and then randomly selected with replacement to generate a set of simulated observations for MPA and Ref with the number of observations matching the actual number of transects surveyed in all MPAs and Refs within the region, respectively. Resampling was repeated 1000 times, producing 1000 sets of observations for MPA and Ref. Then, we took the mean of each set of MPA and Ref observations and calculated the log response ratio with MPA as the numerator and Ref as the denominator, totaling 1000 ratios. If the observed (un-pooled) log response ratio for a region falls above or below the bootstrap 95% CI, the response ratio is significantly positive or negative (i.e., fish biomass is significantly higher inside or outside of the MPA).

Since significant divergence in biomass response ratios (from zero) was detected in at least some years at

most MPAs or regions, we then ran analysis of covariance (ANCOVA) to test whether MPA effects on fish biomass led to increasingly divergent response ratios over time and whether these trends differed by fishing target status. The model terms included target status (targeted vs. non-targeted) as a fixed effect, year as a continuous covariate, target status $\times$ year (interaction), and log biomass ratio as the response variable. Parameter estimates of the slope for biomass ratios over time (year) were tested separately for targeted and non-targeted species to determine whether these groups showed significant trends. These ANCOVA tests were run on each MPA/Ref pair individually and separately on the mean MPA/Ref response ratios across MPAs within each region.

Trajectories of targeted and non-targeted response ratios

We also examined the regional fish responses using a more conservative analytical framework to determine whether MPA effects were still detectable (Appendix S2). We attribute positive effects of MPAs on the overall response of targeted species only when (1) the overall response of targeted species within MPAs and Ref areas exhibits positive divergence from one another since the establishment of the MPA resulting in a significant positive slope of the response ratio of targeted species over time (i.e., [TargetedMPA/TargetedREF]), (2) the response ratios of species targeted by fisheries and species not targeted by fisheries within the MPA (i.e., [TargetedMPA vs. Non-TargetedMPA]) exhibit positive divergence from one another since the establishment of the MPA, and (3) non-targeted species exhibit similar trajectories inside and outside of the MPA.

With this framework, the strongest positive MPA effects are inferred when all three criteria are met. The first criterion of divergence (1) addresses the potential bias of inherent initial differences in population levels between MPA and Ref sites that persist over the monitoring period. Divergence demonstrates the response of populations to MPA establishment over time. Collectively, (2) and (3) act as controls for changes in environmental conditions over time within MPAs and between MPA and Ref sites, under the assumption that the responses of non-targeted species are independent of those of targeted species (for more details, see Appendix S2). Since this framework relies on contrasts between targeted and non-targeted species, we applied this approach only to the fish biomass dataset. Although some invertebrates are targeted (e.g., abalone, spiny lobster, sea cucumbers), there are no comparable non-targeted taxa for this analysis.

All statistical analyses were conducted in SAS Enterprise Guide (SAS Institute Inc. 2013, version 7.1,

Procedure: GLM) and R studio (version 2022.7.1.554; Function: t.test).

RESULTS

Targeted and non-targeted fish biomass response

Average response ratios

We found that 16 of the 20 MPAs (80%) had positive response ratios for targeted species averaged across years, and almost all of these positive responses were significant (12 out of 16), indicating higher biomass on average inside the MPAs ($p < 0.05$, Figure 2, Appendix S3: Table S1). Only four MPAs exhibited negative response ratios (two in the North Coast and one each in the Central and South Coasts) with two of them exhibiting significant negative responses ($p < 0.05$). For non-targeted species, the distribution of positive and negative response ratios was random, with 12 MPAs showing positive responses and 8 MPAs showing negative responses, and about half of these responses were significant ($p < 0.05$).

Relative change in species responses

We explored the magnitude of the difference between MPA and Ref responses (slopes) using Hedges' g . In each region except for the North Coast, the biomass of targeted species increased more (or decreased less) over time inside MPAs than that of the non-targeted species (Figure 3).

Trajectories of targeted and non-targeted response ratios

We did not observe strong and consistent MPA effects in every region using our more conservative framework (Figure 4; Appendix S2). Only two regions (South Coast and Northern Channel Islands) exhibited a significant positive slope in biomass response ratios for targeted species (criterion 1, Figure 4B; Appendix S3: Table S2). Although biomass ratios increased significantly over time inside MPAs for targeted species, none of the regions showed a significant target status by year interaction (i.e., biomass ratio trends for targeted species did not diverge in a positive direction from trends in non-targeted species; criterion 2; Figure 4B; Appendix S3: Table S3), with the exception of the Northern Channel Islands where the interaction between year and target status was marginally non-

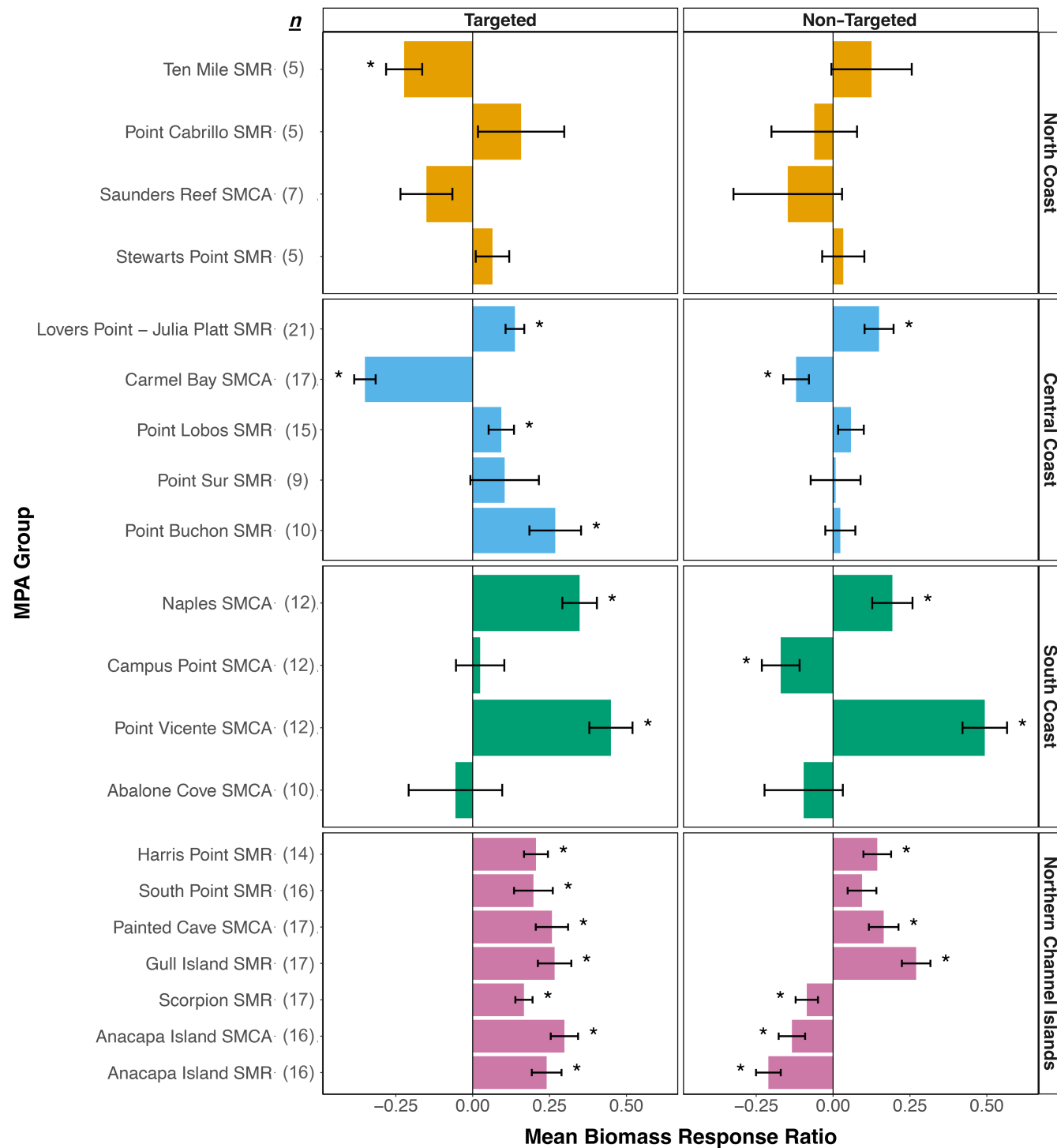


FIGURE 2 Mean biomass response ratio ($\log_{10}(\text{Biomass}_{\text{MPA}}/\text{Biomass}_{\text{Ref}})$) over time with SE for targeted and non-targeted fishes. Biomass is represented as kilograms per 60 m². The responses are grouped by marine protected area (MPA) within each region (n is the number of sampling years for each MPA), as indicated by the values in parentheses. Asterisks denote significance from one-sample t tests ($p < 0.05$). One-sample t test statistics are found in Appendix S3: Table S1.

significant ($n = 36$, $p = 0.08$). In three of the four regions, the non-targeted species biomass inside the MPAs tracked biomass in the Ref areas (Figure 4A) resulting in a non-significant slope of their response ratios (Figure 4B),

fulfilling criterion 3. The one regional exception was the South Coast, where the positive response ratio slope (Figure 4B) reflected a negative trajectory in the Ref areas, while the biomass trajectory remained relatively constant in

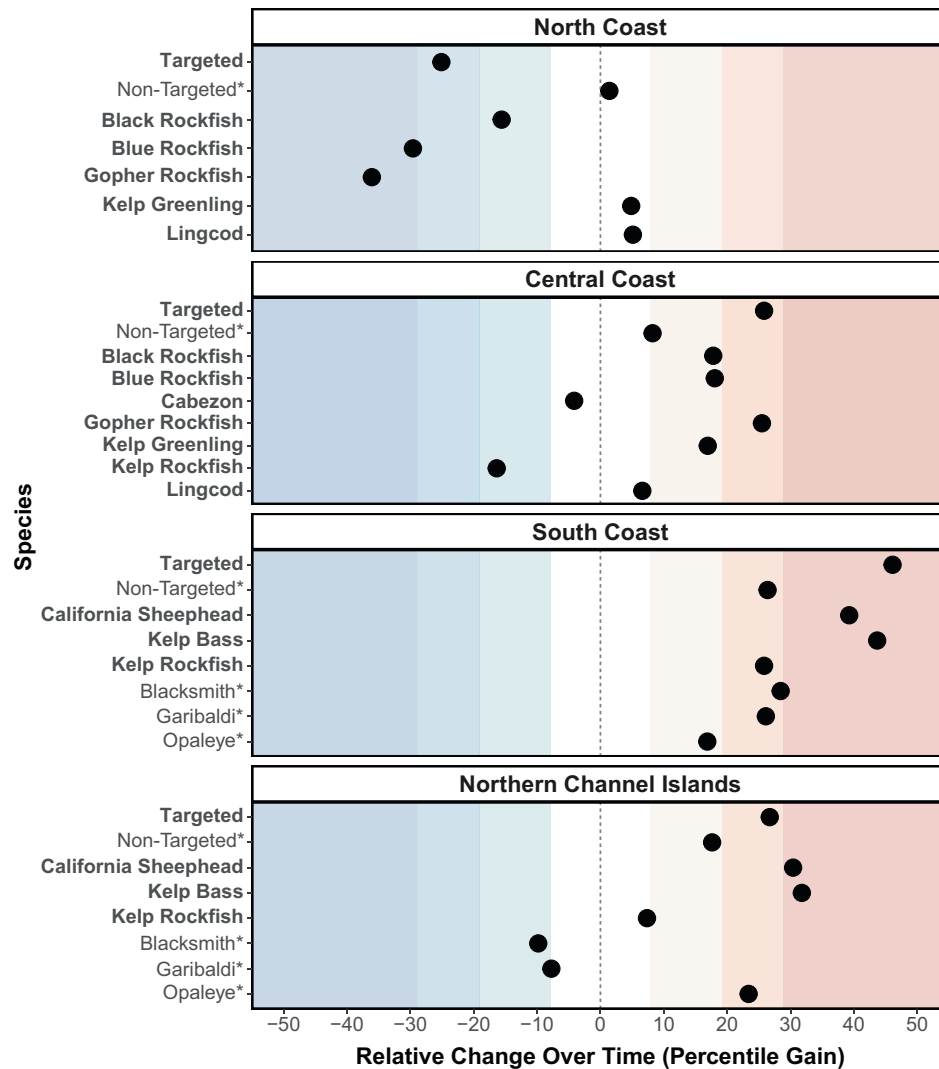


FIGURE 3 Relative change in species responses: relative change in fish biomass inside and outside of marine protected areas (MPAs) over time for each region. Relative change (slope) for MPAs and reference sites were calculated using Hedges' g and expressed as percentile gain. Positive values indicate that average change in biomass over time (slope) for a group in MPA sites is higher than average slope in reference sites. Negative values indicate that the slope in reference sites is higher than average slope in MPA sites. An asterisk after the name or grouping indicates species that are not targeted by fishing. Shading indicates the magnitude of the effect size (or difference between MPA and references) with small, medium, and large effects represented by successively darker shading.

the MPAs. The North Coast met criterion 3, but not 1 or 2; the Central Coast met criterion 3, but not 1 or 2; the South Coast met criterion 1, but not 2 or 3; and the Northern Channel Islands met criteria 1 and 3 and marginally met criterion 2.

Focal fish species biomass responses

Average response ratios

Here, we explore the magnitude of the difference between MPA and Ref responses for each focal fish species (Figure 5A; Appendix S3: Table S4). In all regions

where they occur abundantly, two targeted species, California Sheephead (*Bodianus pulcher*) and Kelp Bass (*Paralabrax clathratus*), and the non-targeted Blacksmith (*Chromis punctipinnis*) showed the greatest average biomass response ratios across years (all significantly greater than zero, $p \leq 0.02$). In the North Coast, targeted Blue Rockfish (*Sebastes mystinus*) mean biomass ratios were negative (higher biomass in Ref areas than in MPAs over time), but not significantly so ($n = 22$, $p = 0.72$). In the Central Coast, Cabezon (*Scorpaenichthys marmoratus*), Gopher Rockfish (*Sebastes carnatus*), and Kelp Rockfish (*Sebastes atrovirens*) (all targeted) were significantly more abundant in MPAs than in Ref areas ($n = 72$, $p < 0.03$), while targeted Black Rockfish (*Sebastes melanops*) and

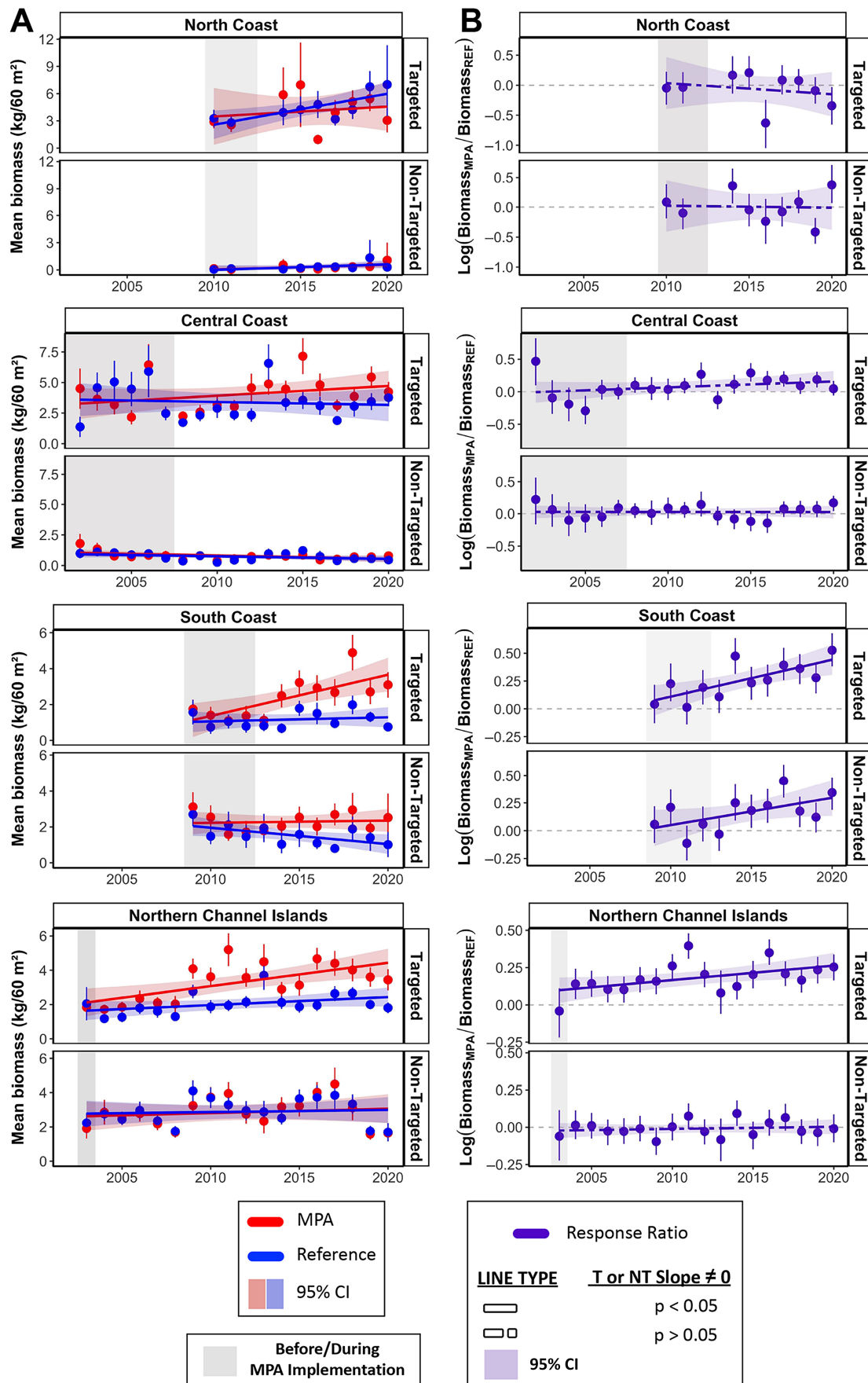


FIGURE 4 Legend on next page.

Kelp Greenling (*Hexagrammos decagrammus*) had higher biomass in Ref areas; however, the mean biomass response ratios was not significantly negative ($n = 72$, $p > 0.07$).

Relative change in species responses

In each region except for the North Coast, the biomass of each focal fish species is increasing more (or decreasing less) over time inside MPAs with the exception of Kelp Rockfish in the Central Coast and Blacksmith and Garibaldi (*Hypsypops rubicundus*) in the Northern Channel Islands (Figure 3). These three species are decreasing more (or increasing less) over time inside MPAs.

Focal invertebrate and algae density responses

Average response ratios

We summarized response ratios for focal invertebrates and algae by averaging over the entire time series at each MPA (Figure 5B; Appendix S3: Table S5) and found very strong regional differences. In the North Coast, red urchins (*Mesocentrotus franciscanus*), purple urchins (*Strongylocentrotus purpuratus*), and bull kelp (*Nereocystis luetkeana*) had higher densities on average in MPAs than in Ref areas over the course of the time series, albeit their responses were not significant ($n = 22$, $p > 0.05$). In the Central Coast, all focal invertebrates except red abalone (*Haliotis rufescens*) had negative response ratios, indicating greater density in Ref areas than in MPAs. Red abalone and red urchins were the only two species in the Central Coast that showed significant positive and negative responses, respectively ($n = 72$, $p < 0.04$). In the South Coast, all focal invertebrates had higher densities in MPAs than in Ref areas with the responses of gorgonians (*Gorgonian* spp.),

sunflower stars (*Pycnopodia helianthoides*), and both urchin species being significant ($n = 46$, $p < 0.04$). The Northern Channel Islands region showed similar responses to the South Coast with the exception of red urchin ($n = 113$, $p = 0.29$), purple urchin ($n = 113$, $p < 0.001$), and gorgonian species ($n = 113$, $p = 0.002$) having higher densities in Ref areas. Most of the species that exhibited significantly strong responses ($p < 0.001$) inside MPAs in the South Coast and Northern Channel Islands regions are targeted (i.e., California spiny lobster [*Panulirus interruptus*], red urchins, and sea cucumbers [*Apostichopus californicus/parvimensis*]).

Relative change in species responses

The density of abalones in the North and Central Coasts and sunflower stars in the Central Coast is increasing more (or decreasing less) over time inside MPAs (Figure 6). Spiny lobsters and sea cucumbers have similar responses in the South Coast and Northern Channel Islands. Purple and red urchins are notably increasing more outside of MPAs in all of the regions, with the exception of purple urchins in the North Coast. The distinct patterns in the response ratios over time for California spiny lobster (positive), purple urchins (negative), and giant kelp (positive) may indicate trophic cascades.

Effect of MPA attributes on performance

Geography (latitude) has a clear and strong effect on the MPA performance (measured by average response ratio for targeted fishes) (Figure 7A), with greater MPA response in the South Coast and Northern Channel Islands as noted above. The Central Coast MPAs showed intermediate responses, with the weakest responses in the North Coast. We found a trend of greater MPA performance with increasing MPA size (in square meters), but this was not significant and variability was high

FIGURE 4 Responses of targeted and non-targeted fishes by region. (A) Trajectories of average biomass (in kilograms per 60 m²) of targeted and non-targeted fishes within marine protected areas (MPAs) and outside (reference, Ref) sites by region. (B) Log response ratios ($\log_{10}(\text{Biomass}_{\text{MPA}}/\text{Biomass}_{\text{Ref}})$) for targeted and non-targeted fish groups by region. In both (A) and (B), each data point represents average response across the region (biomass within MPA or Ref in [A] and biomass response ratio in [B]). Error bars on each data point are 95% CIs (based on individual sites as replicates in [A] and based on bootstrap resampling of individual transects in [B]). Data points are fitted with a regression line (red line = MPA and blue line = Ref in [A] and purple in [B]). Colored shading represents 95% CI for the slope of each respective regression line (red = MPA and blue = Ref in [A] and purple in [B]). Gray-shaded region represents data collected before or during MPA implementation. Line type in (B) represents linear regression results examining whether response of each target status over time (slope) is significantly different from zero. A significant slope departure from zero ($p < 0.05$) is represented as a solid line. A non-significant departure ($p > 0.05$) is represented as a dashed line. Analysis of covariance statistics are found in Appendix S3: Tables S1–S3.

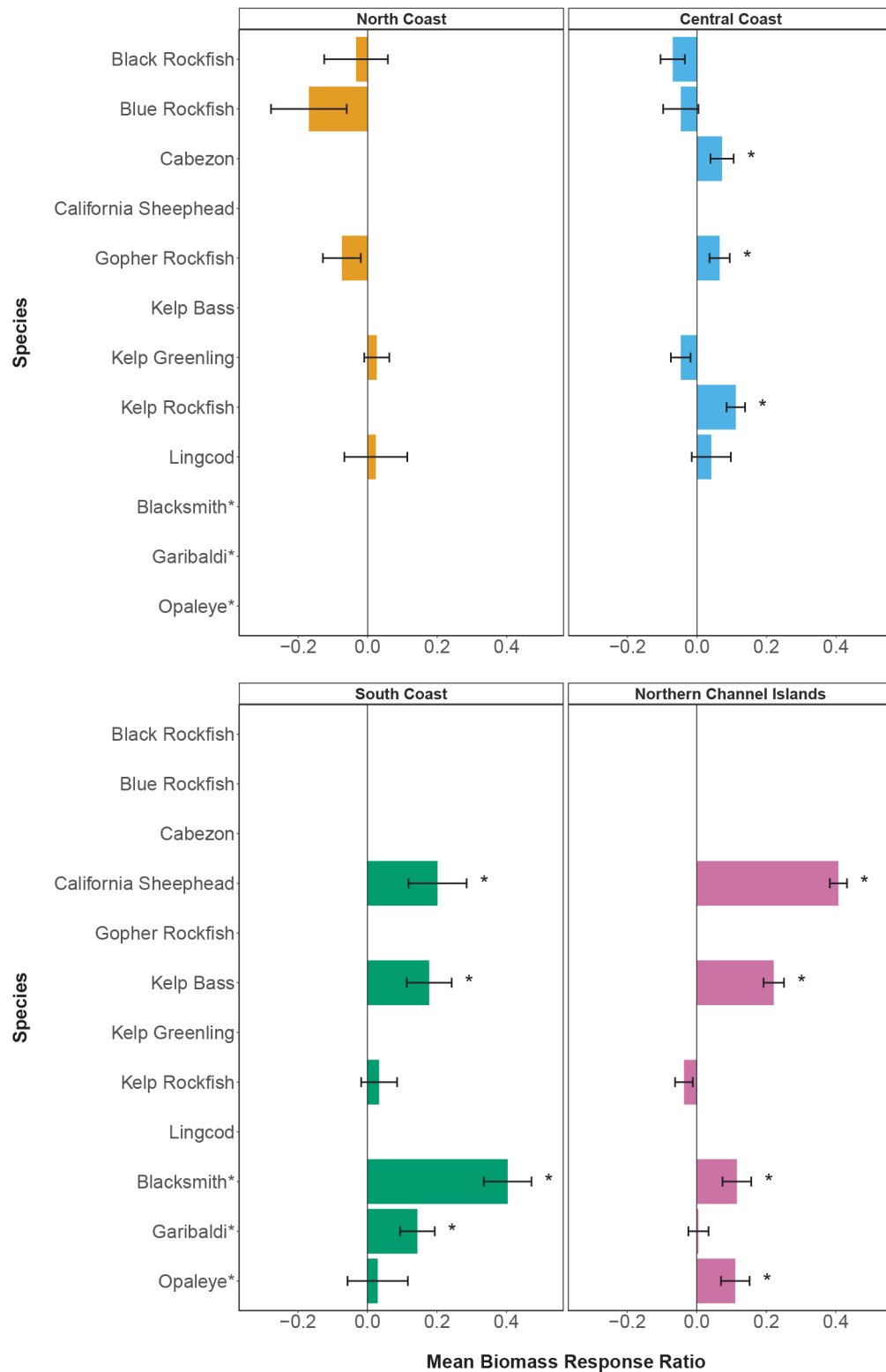


FIGURE 5 Mean log response ratio over time with SE for each focal species: (A) fish ($\log_{10}(\text{Biomass}_{\text{MPA}}/\text{Biomass}_{\text{Ref}})$) and (B) invertebrate and algae ($\log(\text{Density}_{\text{MPA}}/\text{Density}_{\text{Ref}})$). Biomass is represented as kilograms per 60 m², and density is represented as the number of individuals per 60 m². The responses are grouped by species within each region. Sample size, n , is the same for all species within a region (n = total number of sampling years combined across marine protected areas [MPAs] within a region): North Coast (n = 22), Central Coast (n = 72), South Coast (n = 46), and North Channel Islands (n = 113). In panel A, an asterisk after the name indicates fish species that are not targeted by fishing. In both panels, an asterisk above the bar denotes significance from one-sample *t* tests ($p < 0.05$). One-sample *t* test statistics in Appendix S3: Tables S4 and S5.

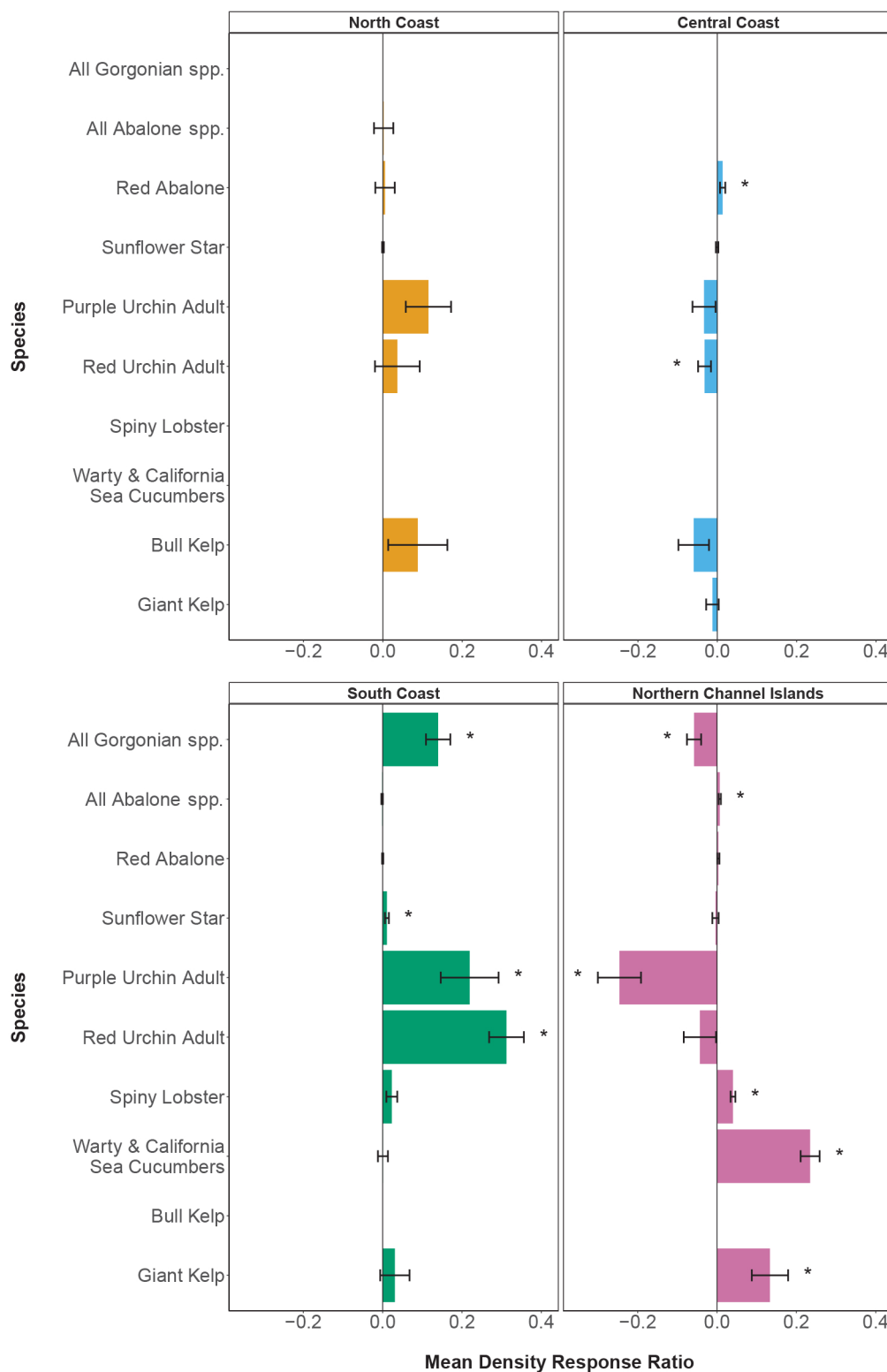


FIGURE 5 (Continued)

(Figure 7B, Appendix S3: Table S6). This lack of a strong MPA size effect was not due to systematic differences among regions, as MPAs of all sizes occur in each region. MPA responses declined with increasing habitat

diversity, habitat richness, and amount of rock reef in MPAs (Figure 7C,D,F). We saw no clear relationship between MPA performance and distance to the nearest port (Figure 7E).

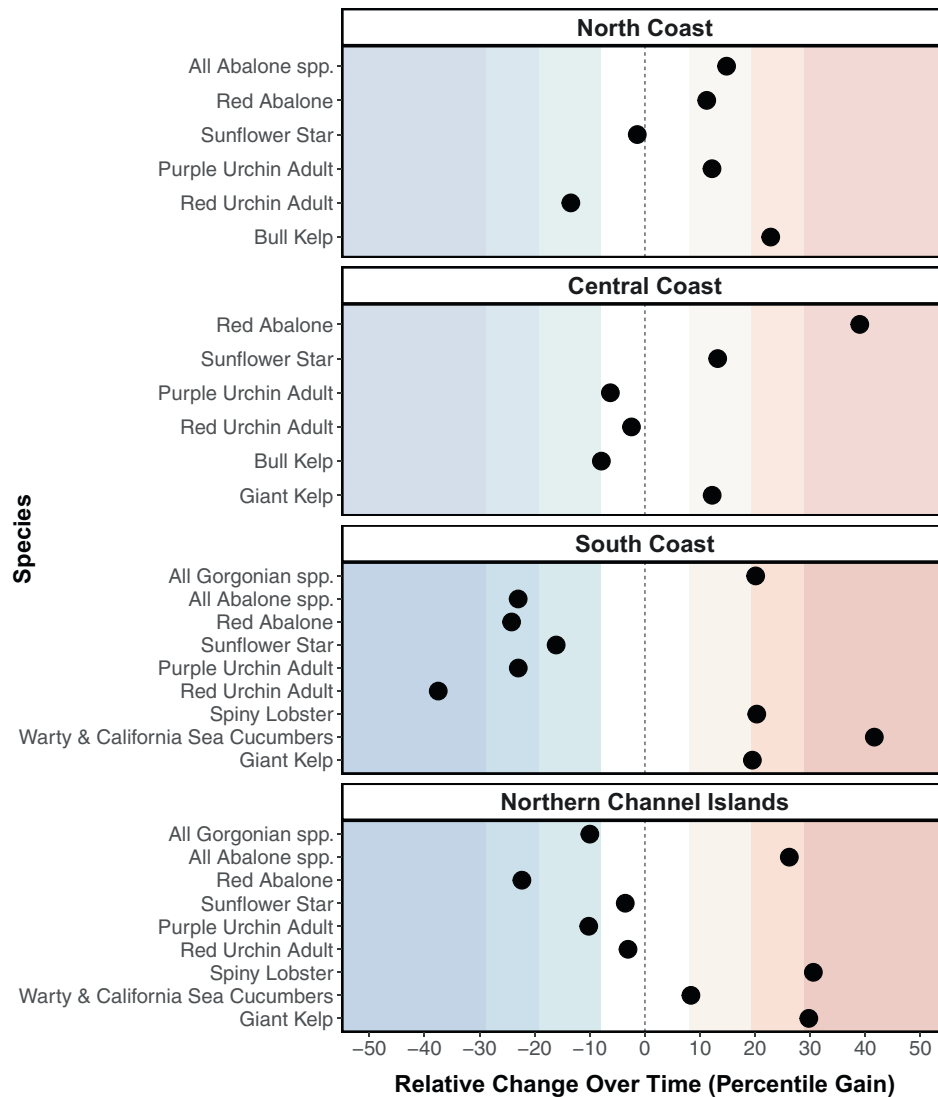


FIGURE 6 Relative change in species responses: relative change in focal invertebrate and algae density inside and outside of marine protected areas (MPAs) over time for each region. Relative change (slope) for MPAs and reference sites (Refs) were calculated using Hedges' g and expressed as percentile gain. Positive values indicate that average change in biomass over time (slope) for a group in MPA sites is higher than average slope in Refs. Negative values indicate that the slope in Refs is higher than average slope in MPA sites. Shading indicates the magnitude of the effect size (or difference between MPA and Refs) with small, medium, and large effects represented by successively darker shading.

DISCUSSION

MPA performance varies across taxa and region

We found that population responses to protection in MPAs across the network varied greatly whether evaluated as combined taxa (e.g., targeted and non-targeted by fisheries), or individual focal species, or at a single MPA or regional scales. Despite this variability, some general patterns emerged using analytical methods that increased in complexity including effect sizes, average response ratios, and trajectories of targeted/non-targeted response ratios

as measures of MPA performance. For both fish and invertebrate species, the strongest positive responses in MPAs occurred for those species that are heavily targeted by fisheries and at MPAs in both of the southern regions of the network (South Coast and Northern Channel Islands), where fishing pressure is greatest (Smith et al., 2023).

We detected significant positive effects on targeted species in 12 of the 20 MPAs evaluated, supporting the general consensus that MPAs can have positive consequences for populations and communities (Ferreira et al., 2022; Giakoumi et al., 2017; Grorud-Colvert et al., 2014; Lester et al., 2009; Ohayon et al., 2021;

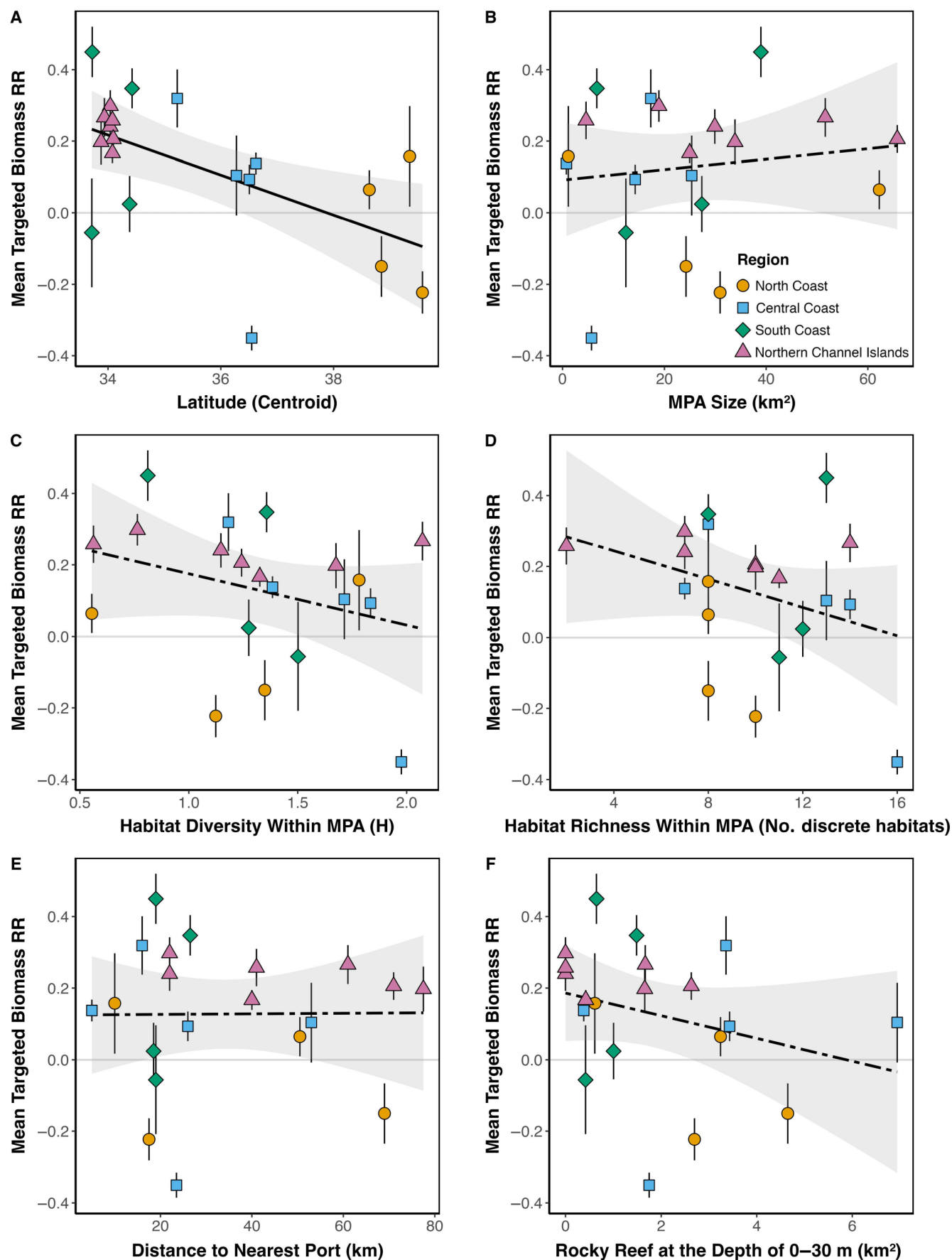


FIGURE 7 Legend on next page.

Vandeperre et al., 2011). However, these effects are not uniform across a large network, with significant negative effects detected in two MPAs and no significant effects in another 6 of the 20 MPAs evaluated (Figure 2). The marked geographic variation in population responses that we detected likely reflects the separate and combined effects of regional differences in design attributes of individual MPAs (e.g., size, levels of protection and compliance, time since establishment), the network (e.g., regional differences in larval connectivity), attributes of species (e.g., targeted vs. non-targeted, faster population growth of warm water affinity species), regional differences in fishing mortality outside of MPAs, environmental differences (e.g., temperature), including the impacts of the 2014–2016 marine heatwave, and the duration of time series that underpin the statistical power of our analyses. In some MPAs, positive effects on targeted species (e.g., California Sheephead, spiny lobster) might have had negative effects on non-targeted species (purple urchins), resulting in cascading indirect positive effects on other species (giant kelp; e.g., Northern Channel Islands; Caselle et al., 2018; Eisaguirre et al., 2020; Kumagai et al., 2024; Pinnegar et al., 2000).

We tested a suite of simple MPA attributes previously shown to influence MPA performance such as MPA size, habitat characteristics, and distance from port (a proxy for fishing effort) but found little clear effect beyond the strong effect of geography. Unlike Ziegler et al. (2023), we did not test the effect of MPA age which has been shown to be an important attribute in prior studies (Claudet et al., 2008; Edgar et al., 2014; Friedlander et al., 2017). In California, MPAs were implemented region by region, each at different times, confounding the effects of California's well known biogeography with MPA age. Similar to many MPA studies (e.g., Claudet et al., 2008; Edgar et al., 2014; Halpern, 2003; Vandeperre et al., 2011), we did find a positive, although non-significant, effect of MPA size on MPA performance. The effects of habitat attributes on MPA performance have rarely been tested (but see Jainese et al., 2024; Smith et al., 2024). In California, we have access to comprehensive benthic habitat data in all MPAs in the network. Habitat diversity and richness, as well as amount of rock habitat within an MPA, were non-significantly but negatively correlated with MPA performance, contrary to our predictions that more habitat types and greater area of rock inside the MPA might increase the biomass of fishes

(similarly, see Bosch et al., 2022; Pondella et al., 2019). A key variable that should be further considered is fishing mortality at scales relevant to the MPAs. Interestingly, we found no relationship in MPA response with distance to nearest port across all regions, but when comparing MPA responses by the identity of the nearest port, we found that MPAs closest to the four southernmost ports (Morro Bay, Santa Barbara, Channel Islands, and San Pedro) showed greater positive MPA responses than those nearest Ft. Bragg, Bodega Bay, and Monterey in northern and central California (Carr et al., 2021). Studies have shown that the level of exploitation can markedly impact targeted populations and, in turn, affect MPA efficacy (Nickols et al., 2019; White et al., 2010, 2013). Our strongest positive responses to MPAs occurred in the South Coast and Northern Channel Islands—the regions with the highest coastal human population density in the state and greatest fishing pressure (Bennett et al., 2004; Free et al., 2022; Rees et al., 2021; Smith et al., 2023). Unfortunately, spatial data on fishing mortality at the scale of individual MPA-Ref pairs are lacking in California, as in most parts of the world. Despite the strong positive MPA effects in these southern regions, the responses did vary from MPA to MPA and the variation may be attributed to other MPA characteristics mentioned above in addition to the fact that the fishing effort within the Southern California region, although high, is spatially variable itself (Zellmer et al., 2018).

The large variation we observed in species responses to MPA establishment—both for the same species across different MPAs and for the different species within the same MPA—is not surprising and is consistent with findings from other large-scale MPA networks. This spatial variation reflects a combination of ecological, environmental, and anthropogenic factors. Again, species targeted by commercial and/or recreational fisheries often respond more strongly to protection, but the magnitude and rate of these responses depend heavily on pre-existing fishing pressure and fishing mortality rates (Jaco & Steele, 2020; Nickols et al., 2019; White et al., 2013). Furthermore, the effect of reduced fishing pressure interacts with species' natural mortality rates—species with lower natural mortality often show more pronounced biomass recovery after protection (Kaplan et al., 2019). Life history traits also modulate response; species with smaller home ranges relative to MPA size are more likely to benefit from protection due to greater residency within

FIGURE 7 Relationship between the mean response ratios of targeted species and marine protected area (MPA) attributes: (A) latitude, (B) MPA size, (C) habitat diversity in MPA, (D) habitat richness in MPA, (E) distance to nearest port, and (F) area of rocky reef from 0 to 30 m depth. Slopes are linear regressions with dashed slopes not significantly different from zero. Regression statistics are found in Appendix S3: Table S6.

reserves (Grüss et al., 2011). However, even among the relatively site-attached rocky reef fishes in our study, home range size varies considerably (Freiwald, 2012).

Crucially, these biological and ecological factors are overlaid on a spatial mosaic of environmental conditions and human use that differ across California's MPAs. Previous studies in the region have consistently found high spatial heterogeneity in MPA responses—Hamilton et al. (2010) documented strong biogeographic structuring of fish assemblages in the Channel Islands, and Caselle et al. (2015) showed that biomass trajectories of targeted species differed across islands likely due to variation in oceanography, fishing pressure, and reserve design. Similarly, Starr et al. (2015) reported slow and variable fishery responses across central California MPAs, noting that benefits were most evident in the oldest and best-enforced reserves. Comparable patterns of spatially variable responses have been documented in large-scale MPA networks elsewhere, including the Great Barrier Reef (Emslie et al., 2015; McCook et al., 2010), the Mediterranean Sea (Guidetti et al., 2014), and southeast Australia (Knott et al., 2021). Collectively, these studies support the conclusion that regional and local context—ecological, physical, and anthropogenic—plays a critical role in shaping MPA performance and that variation in outcomes is both expected and informative for adaptive management.

Given this spatial and ecological variability, appropriate methodological frameworks are essential to disentangle MPA effects from background environmental variation. One solution has been the application of time series to compare average values before and after MPA implementation (i.e., before-after-control-impact [BACI]; Ojeda-Martínez et al., 2011; Osenberg et al., 2011; Sciberras et al., 2013; Tissot et al., 2004). However, by focusing on average conditions before and after MPA implementation, many BACI designs may overlook changes in the trajectory or rate of change (i.e., trends) triggered by the intervention (Wauchope et al., 2021). Moreover, whereas the California MPA network was scientifically designed, coordinated monitoring prior to implementation was limited, making it difficult to apply traditional before–after approaches and further highlighting the need for alternative frameworks. In response, we applied an alternative approach that does not require pre-implementation time series but instead evaluates divergence between MPA and non-MPA sites, as well as between targeted and non-targeted species, subsequent to MPA establishment to enable rigorous attribution of MPA effects (Appendix S2).

As expected, aggregated populations targeted by fisheries varied regionally in the extent to which they fulfilled all three criteria for MPA effects in the California

network. North and Central Coasts fulfilled only the third criterion, indicating that non-targeted species responses did not diverge between inside and outside of the MPAs. By contrast, South Coast exhibited positive divergence in biomass trajectories for targeted species between inside and outside of MPAs; however, this divergence was not significantly different from that observed for non-targeted species (e.g., South Coast; Figure 4; Appendix S3: Tables S2 and S3). Only Northern Channel Islands met all three criteria (Figure 4; Appendix S3: Tables S2 and S3); however, the targeted and non-targeted interaction was only marginally significant ($n = 34$, $p = 0.0855$).

While the framework provides a conservative approach to attributing MPA effects, it is not without its limitations, which stem from key assumptions underlying its design. The framework assumes that population trajectories inside and outside of MPAs are still diverging, as populations have not yet reached alternative equilibria reflecting relative mortality rates. It also assumes that fishing mortality rates and trajectories in reference areas represent general conditions outside of MPAs, though this assumption is difficult to test without spatially relevant fishing mortality data across California's MPA network. Additionally, the framework presumes that non-targeted species' trajectories remain independent of targeted species, despite potential ecological interactions, such as predator–prey dynamics, influenced by changes in targeted species' abundance. Given these assumptions, the framework we present may be most useful in the early period following MPA implementation, a period where monitoring is often most robust and funded, and before strong trophic interactions develop and populations reach asymptotic level.

The timing and duration of monitoring may have also contributed to some of the regional differences in the detection of population responses. In combination, the shorter monitoring periods, beginning in 2010 and 2012, and the 2014–2016 marine heatwave likely impaired our ability to detect population responses across the North Coast MPAs. These were the last MPAs to be established, and the 2014–2016 marine heatwave disproportionately impacted the North Coast, devastating kelp forest ecosystems (Beas-Luna et al., 2020; McPherson et al., 2021; Rogers-Bennett & Catton, 2019). The short time series in that region, which began only one year prior to the MHW, prevented us from distinguishing the effects of the MHW from the potential development of any population response to the MPA.

Regional differences across MPA networks may also stem from varying levels of ecological connectivity among MPAs and the resulting replacement capacity of species' metapopulations. For instance, recent analyses

of the contribution of California's MPA network to the replacement capacity of select kelp forest fishes found that the network had a far greater consequence for kelp bass (*P. clathratus*) than three rockfishes of colder water affinity (genus *Stegastes*) (Yeager et al., 2023). These results underscore the importance of integrating spatially explicit network models (i.e., demographic-based population connectivity) and empirical assessments of population and community responses for evaluations of MPA network performance. Spatially explicit models that estimate network-wide patterns of population and community connectivity can estimate network-wide population and community responses and evaluate the relative contribution of individual MPAs to network performance. But these models are most useful when informed by empirically derived species–habitat relationships to generate species distribution models, the size structure of populations (including recruitment) from which patterns of larval production can be derived, and other demographic and community-level metrics (e.g., composition and diversity) gleaned from monitoring programs (e.g., Botsford et al., 2009; Burgess et al., 2014; Dedrick et al., 2021; Garavelli et al., 2018; Johnson et al., 2018). While the data generated by our monitoring program allow for population response assessments, it is also producing the distributional and demographic data that are being applied to models of network performance. The design of monitoring studies should be conducted in concert with network models to ensure such studies are informing those network analyses.

Regionally tailored management of a MPA network may be more effective and nimble

While large MPA networks can offer advantages over a single, large MPA, they also pose unique challenges for management and monitoring. For example, in California's North Coast, access to kelp forests for SCUBA surveys is limited due to extreme ocean conditions, resulting in less robust time series than in other regions. These access challenges are not exclusive to any particular type of MPA, but when monitoring must be coordinated across a geographically extensive network, such challenges can become more pronounced.

Networks that span a large geographic range, such as California's, must also contend with regional differences in the magnitude and types of fishing, fisheries management, physical access to MPAs, human population densities, stakeholder interests, and other environmental and social factors. Although one of the most important design attributes of an MPA network is its integration across the full extent of a governing body's coastal

waters (Grorud-Colvert et al., 2014), our results strongly suggest that tailoring management to regional ecological and human contexts may improve efficiency, responsiveness, and overall effectiveness.

Those managing large MPA networks elsewhere might consider establishing regional sub-networks under a central managing agency, delegating certain responsibilities—such as levels and types of monitoring, enforcement strategies, outreach, partnerships, and resource allocation—to allow for customization that reflects regional conditions and priorities, thereby strengthening the cohesion and success of the broader network.

AUTHOR CONTRIBUTIONS

All authors contributed to the manuscript conception and design. Mark H. Carr, Jennifer E. Caselle, Daniel P. Malone, Kathryn Koehn, Avrey Parsons-Field, and Daniel J. Pondella II collected the data. Daniel P. Malone and Barbara Spiecker analyzed the data. All authors contributed critically to the drafts and gave final approval for publication.

ACKNOWLEDGMENTS

We thank the hundreds of individuals who contributed to data collection for each of the long-term monitoring programs included in this study: kelp forest monitoring groups at UC Santa Barbara and UC Santa Cruz (Partnership for Interdisciplinary Studies of Coastal Oceans, PISCO), the Vantuna Research Group at Occidental College, and Cal Poly Humboldt. This research was funded, in part, by the California Ocean Protection Council who provided support for long-term monitoring through California Sea Grant (R/MPA-43, R/MPA-49A), the Gordon and Betty Moore Foundation, and the David and Lucile Packard Foundation. Jennifer E. Caselle, Mark H. Carr, Daniel P. Malone, Avrey Parsons-Field, and Barbara Spiecker were supported, in part, by the David and Lucile Packard Foundation through the Partnership for Interdisciplinary Studies of Coastal Oceans (PISCO). This is PISCO publication number 550.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data (Carr et al., 2023) are available in DataONE at https://doi.org/10.25494/P6/MLPA_kelpforest.5.

ORCID

Barbara Spiecker  <https://orcid.org/0000-0002-6895-9424>

Mark H. Carr  <https://orcid.org/0000-0001-9644-7680>

Daniel P. Malone  <https://orcid.org/0000-0002-1758-7026>

REFERENCES

- Addison, P. F. E., L. B. Flander, and C. N. Cook. 2017. "Towards Quantitative Condition Assessment of Biodiversity Outcomes: Insights from Australian Marine Protected Areas." *Journal of Environmental Management* 198: 183–191.
- Ahmadia, G. N., L. Glew, M. Provost, D. Gill, N. I. Hidayat, S. Mangubhai, Purwanto, and H. E. Fox. 2015. "Integrating Impact Evaluation in the Design and Implementation of Monitoring Marine Protected Areas." *Philosophical Transactions of the Royal Society B: Biological Sciences* 370: 20140275.
- Allen, L. G., D. J. Pondella, and M. H. Horn. 2006. *The Ecology of Marine Fishes: California and Adjacent Waters*. Berkeley, CA: University of California Press.
- Baetscher, D. S., E. C. Anderson, E. A. Gilbert-Horvath, D. P. Malone, E. T. Saarman, M. H. Carr, and J. C. Garza. 2019. "Dispersal of a Nearshore Marine Fish Connects Marine Reserves and Adjacent Fished Areas along an Open Coast." *Molecular Ecology* 28: 1611–23.
- Ballantine, B. 2014. "Fifty Years On: Lessons from Marine Reserves in New Zealand and Principles for a Worldwide Network." *Biological Conservation* 176: 297–307.
- Batista, M. I., and H. N. Cabral. 2016. "An Overview of Marine Protected Areas in SW Europe: Factors Contributing to Their Management Effectiveness." *Ocean & Coastal Management* 132: 15–23.
- Beas-Luna, R., F. Micheli, C. B. Woodson, M. Carr, D. Malone, J. Torre, C. Boch, et al. 2020. "Geographic Variation in Responses of Kelp Forest Communities of the California Current to Recent Climatic Changes." *Global Change Biology* 26: 6457–73.
- Bennett, W., K. Roinestad, L. Rogers-Bennett, L. Kaufman, D. Wilson-Vandenberg, and B. Heneman. 2004. "Inverse Regional Responses to Climate Change and Fishing Intensity by the Recreational Rockfish (*Sebastes* spp.) Fishery in California." *Canadian Journal of Fisheries and Aquatic Sciences* 61: 2499–2510.
- Blanchette, C. A., C. Melissa Miner, P. T. Raimondi, D. Lohse, K. E. K. Heady, and B. R. Broitman. 2008. "Biogeographical Patterns of Rocky Intertidal Communities along the Pacific Coast of North America." *Journal of Biogeography* 35: 1593–1607.
- Boero, F. 2017. *From Marine Protected Areas to MPA Networks. Page Management of Marine Protected Areas: A Network Perspective*. Hoboken, NJ: John Wiley & Sons.
- Boonzaier, L., and D. Pauly. 2016. "Marine Protection Targets: An Updated Assessment of Global Progress." *Oryx* 50: 27–35.
- Bosch, N. E., J. Monk, J. Goetze, S. Wilson, R. C. Babcock, N. Barrett, J. Clough, et al. 2022. "Effects of Human Footprint and Biophysical Factors on the Body-Size Structure of Fished Marine Species." *Conservation Biology* 36(2): e13807.
- Botsford, L. W., J. W. White, M. H. Carr, and J. E. Caselle. 2014. "Marine Protected Area Networks in California, USA." *Advances in Marine Biology* 69: 205–251.
- Botsford, L. W., J. W. White, M. A. Coffroth, C. B. Paris, S. Planes, T. L. Shearer, S. R. Thorrold, and G. Jones. 2009. "Connectivity and Resilience of Coral Reef Metapopulations in Marine Protected Areas: Matching Empirical Efforts to Predictive Needs." *Coral Reefs* 28: 327–337.
- Burgess, S. C., K. J. Nickols, C. D. Griesemer, L. A. Barnett, A. G. Dedrick, E. V. Satterthwaite, L. Yamane, S. G. Morgan, J. W. White, and L. W. Botsford. 2014. "Beyond Connectivity: How Empirical Methods Can Quantify Population Persistence to Improve Marine Protected-Area Design." *Ecological Applications* 24: 257–270.
- California Department of Fish and Wildlife and Ocean Protection Council. 2018. "Marine Protected Area Monitoring Action Plan."
- Carr, M. H., J. E. Caselle, K. Cavanaugh, J. Freiwald, K. Kroeker, D. Pondella, B. Tissot, D. Malone, A. Parsons-Field, and B. Spiecker. 2021. *Monitoring and Evaluation of Kelp Forest Ecosystems in the MLPA Marine Protected Area Network* 1–193. California Sea Grant, Ocean Protection Council, California Department of Fish and Wildlife.
- Carr, M. H., and E. L. Hazen. 2019. "Corridor Ecology: Linking Landscapes for Biodiversity Conservation and Climate Adaptation." In *Ecological Connectivity in the Ocean*, 2nd ed., edited by J. A. Hilty, A. T. H. Keeley, W. Z. Lidicker, and A. M. Merenlender, 216–237. Washington, D.C.: Island Press.
- Carr, M. H., and D. C. Reed. 2016. "Chapter 17: Shallow Rocky Reefs and Kelp Forests." In *Ecosystems of California*, edited by H. Mooney and E. Zavaleta, 311–336. Berkeley, CA: University of California Press.
- Carr, M. H., S. P. Robinson, C. Wahle, G. Davis, S. Kroll, S. Murray, E. J. Schumacker, and M. Williams. 2017. "The Central Importance of Ecological Spatial Connectivity to Effective Coastal Marine Protected Areas and to Meeting the Challenges of Climate Change in the Marine Environment." *Aquatic Conservation: Marine and Freshwater Ecosystems* 27: 6–29.
- Carr, M. H., J. E. Caselle, B. N. Tissot, D. J. Pondella, D. P. Malone, K. D. Koehn, J. T. Claisse, J. P. Williams, and A. Parsons-Field. 2023. "Monitoring and Evaluation of Kelp Forest Ecosystems in the MLPA Marine Protected Area Network." California Ocean Protection Council Data Repository. https://doi.org/10.25494/P6/MLPA_kelpforest.5.
- Caselle, J. E., K. Davis, and L. M. Marks. 2018. "Marine Management Affects the Invasion Success of a Non-Native Species in a Temperate Reef System in California, USA." *Ecology Letters* 21: 43–53.
- Caselle, J. E., A. Rassweiler, S. L. Hamilton, and R. R. Warner. 2015. "Recovery Trajectories of Kelp Forest Animals Are Rapid Yet Spatially Variable across a Network of Temperate Marine Protected Areas." *Scientific Reports* 5: 14102.
- Claudet, J., C. W. Osenberg, L. Benedetti-Cecchi, P. Domenici, J.-A. García-Charton, Á. Pérez-Ruzafa, F. Badalamenti, et al. 2008. "Marine Reserves: Size and Age Do Matter." *Ecology Letters* 11: 481–89.
- Dedrick, A. G., K. A. Catalano, M. R. Stuart, J. W. White, H. R. Montes, and M. L. Pinsky. 2021. "Persistence of a Reef Fish Metapopulation Via Network Connectivity: Theory and Data." *Ecology Letters* 24: 1121–32.
- Edgar, G. J., R. D. Stuart-Smith, T. J. Willis, S. Kininmonth, S. C. Baker, S. Banks, N. S. Barrett, et al. 2014. "Global Conservation Outcomes Depend on Marine Protected Areas with Five Key Features." *Nature* 506: 216–220.
- Eisaguirre, J. H., J. M. Eisaguirre, K. Davis, P. M. Carlson, S. D. Gaines, and J. E. Caselle. 2020. "Trophic Redundancy and Predator Size Class Structure Drive Differences in Kelp Forest Ecosystem Dynamics." *Ecology* 101: e02993.
- Emslie, M. J., M. Logan, D. H. Williamson, A. M. Ayling, M. A. MacNeil, D. Ceccarelli, A. J. Cheal, et al. 2015. "Expectations

- and Outcomes of Reserve Network Performance Following Re-Zoning of the Great Barrier Reef Marine Park.” *Current Biology* 25(8): 983–992.
- Fenberg, P. B., J. E. Caselle, J. Claudet, M. Clemence, S. D. Gaines, J. Antonio García-Charton, E. J. Gonçalves, et al. 2012. “The Science of European Marine Reserves: Status, Efficacy, and Future Needs.” *Marine Policy* 36: 1012–21.
- Ferreira, H. M., R. A. Magris, S. R. Floeter, and C. E. L. Ferreira. 2022. “Drivers of Ecological Effectiveness of Marine Protected Areas: A Meta-Analytic Approach from the Southwestern Atlantic Ocean (Brazil).” *Journal of Environmental Management* 301: 113889.
- Free, C. M., C. Vargas Poulsen, L. F. Bellquist, S. N. Wassermann, and K. L. Oken. 2022. “The CALFISH Database: A Century of California’s Non-Confidential Fisheries Landings and Participation Data.” *Ecological Informatics* 69: 101599.
- Freiwald, J. 2012. “Movement of Adult Temperate Reef Fishes Off the West Coast of North America.” *Canadian Journal of Fisheries and Aquatic Sciences* 69: 1362–74.
- Friedlander, A. M., Y. Golbuu, E. Ballesteros, J. E. Caselle, M. Gouezo, D. Olsudong, and E. Sala. 2017. “Size, Age, and Habitat Determine Effectiveness of Palau’s Marine Protected Areas.” *PLoS One* 12: e0174787.
- Gaines, S. D., C. White, M. H. Carr, and S. R. Palumbi. 2010. “Designing Marine Reserve Networks for both Conservation and Fisheries Management.” *Proceedings of the National Academy of Sciences of the United States of America* 107: 18286–93.
- Garavelli, L., J. W. White, I. Chollett, and L. M. Chérubin. 2018. “Population Models Reveal Unexpected Patterns of Local Persistence despite Widespread Larval Dispersal in a Highly Exploited Species.” *Conservation Letters* 11: e12567.
- Giakoumi, S., C. Scianna, J. Plass-Johnson, F. Micheli, K. Grorud-Colvert, P. Thiriet, J. Claudet, et al. 2017. “Ecological Effects of Full and Partial Protection in the Crowded Mediterranean Sea: A Regional Meta-Analysis.” *Scientific Reports* 7: 8940.
- Gill, D. A., M. B. Mascia, G. N. Ahmadi, L. Glew, S. E. Lester, M. Barnes, I. Craigie, et al. 2017. “Capacity Shortfalls Hinder the Performance of Marine Protected Areas Globally.” *Nature* 543: 665–69.
- Gleason, M., E. Fox, S. Ashcraft, J. Vasques, E. Whiteman, P. Serpa, E. Saarman, et al. 2013. “Designing a Network of Marine Protected Areas in California: Achievements, Costs, Lessons Learned, and Challenges Ahead.” *Ocean & Coastal Management* 74: 90–101.
- Grorud-Colvert, K., J. Claudet, B. N. Tissot, J. E. Caselle, M. H. Carr, J. C. Day, A. M. Friedlander, et al. 2014. “Marine Protected Area Networks: Assessing whether the Whole Is Greater than the Sum of Its Parts.” *PLoS One* 9: e102298.
- Grorud-Colvert, K., J. Sullivan-Stack, C. Roberts, V. Constant, B. Horta, E. Costa, E. P. Pike, et al. 2021. “The MPA Guide: A Framework to Achieve Global Goals for the Ocean.” *Science* 373: 1–10.
- Grüss, A., D. M. Kaplan, S. Guénette, C. M. Roberts, and L. W. Botsford. 2011. “Consequences of Adult and Juvenile Movement for Marine Protected Areas.” *Biological Conservation* 144: 692–702.
- Guidetti, P., P. Baiata, E. Ballesteros, A. Di Franco, B. Hereu, E. Macpherson, F. Micheli, et al. 2014. “Large-Scale Assessment of Mediterranean Marine Protected Areas Effects on Fish Assemblages.” *PLoS One* 9(4): e91841.
- Hall-Arber, M., S. Murray, L. Aylesworth, M. Carr, J. Field, K. Grorud-Colvert, R. Martone, K. Nickols, E. Saarman, and S. Wertz. 2021. *Scientific Guidance in Support of MPA Decadal Management Reviews – Ocean Science Trust*. Ocean Protection Council Science Advisory Team and California Ocean Science Trust.
- Halpern, B. S. 2003. “The Impact of Marine Reserves: Do Reserves Work and Does Reserve Size Matter?” *Ecological Applications* 13: 117–137.
- Hamilton, S. L., J. E. Caselle, D. P. Malone, and M. H. Carr. 2010. “Incorporating Biogeography into Evaluations of the Channel Islands Marine Reserve Network.” *Proceedings of the National Academy of Sciences of the United States of America* 107: 18272–77.
- Harrison, H. B., D. H. Williamson, R. D. Evans, G. R. Almany, S. R. Thorrold, G. R. Russ, K. A. Feldheim, et al. 2012. “Larval Export from Marine Reserves and the Recruitment Benefit for Fish and Fisheries.” *Current Biology* 22: 1023–28.
- IUCN World Commission on Protected Areas (IUCN-WCPA). 2008. *Establishing Marine Protected Area Networks—Making It Happen*. Washington, DC: IUCN-WCPA, National Oceanic and Atmospheric Administration, and the Nature Conservancy.
- Jaco, E. M., and M. A. Steele. 2020. “Pre-Closure Fishing Pressure Predicts Effects of Marine Protected Areas.” *Journal of Applied Ecology* 57: 229–240.
- Jainese, C., P. M. Carlson, K. Seeto, L. McNeill, K. Siverston, and J. E. Caselle. 2024. “Understanding Organism-Habitat Relationships and Critically Evaluating Reference Areas Is Key to Marine Protected Area Assessment.” *Marine Ecology Progress Series* 751: 115131.
- Jessen, S., L. E. Morgan, J. E. Bezaury-Creel, A. Barron, R. Govender, E. P. Pike, V. R. Saccomanno, and R. A. Moffitt. 2017. “Measuring MPAs in Continental North America: How Well Protected Are the Ocean Estates of Canada, Mexico, and the USA?” *Frontiers in Marine Science* 4:1–11.
- Johnson, D. W., M. R. Christie, T. J. Pusack, C. D. Stallings, and M. A. Hixon. 2018. “Integrating Larval Connectivity with Local Demography Reveals Regional Dynamics of a Marine Metapopulation.” *Ecology* 99: 1419–29.
- Jones, K. R., C. J. Klein, H. S. Grantham, H. P. Possingham, B. S. Halpern, N. D. Burgess, S. H. M. Butchart, et al. 2020. “Area Requirements to Safeguard Earth’s Marine Species.” *One Earth* 2: 188–196.
- Kaplan, K. A., L. Yamane, L. W. Botsford, M. L. Baskett, A. Hastings, S. Worden, and J. W. White. 2019. “Setting Expected Timelines of Fished Population Recovery for the Adaptive Management of a Marine Protected Area Network.” *Ecological Applications* 29: e01949.
- Knott, N. A., J. Williams, D. Harasti, H. A. Malcolm, M. A. Coleman, B. P. Kelaher, M. J. Rees, A. Schultz, and A. Jordan. 2021. “A Coherent, Representative, and Bioregional Marine Reserve Network Shows Consistent Change in Rocky Reef Fish Assemblages.” *Ecosphere* 12: e03447.
- Krumhansl, K. A., D. K. Okamoto, A. Rassweiler, M. Novak, J. J. Bolton, K. C. Cavanaugh, S. D. Connell, et al. 2016. “Global Patterns of Kelp Forest Change over the Past Half-Century.” *Proceedings of the National Academy of Sciences of the United States of America* 113: 13785–90.
- Kumagai, J. A., M. C. Goodman, J. C. Villaseñor-Derbez, D. S. Schoeman, K. C. Cavanaugh, T. W. Bell, F. Micheli, G. De

- Leo, and N. Arafeh-Dalmau. 2024. "Marine Protected Areas that Preserve Trophic Cascades Promote Resilience of Kelp Forests to Marine Heatwaves." *Global Change Biology* 30(12): e17620.
- Lester, S., B. Halpern, K. Grorud-Colvert, J. Lubchenco, B. Ruttenberg, S. Gaines, S. Aíramé, and R. Warner. 2009. "Biological Effects within No-Take Marine Reserves: A Global Synthesis." *Marine Ecology Progress Series* 384: 33–46.
- Lowry, G. K., A. T. White, and P. Christie. 2009. "Scaling Up to Networks of Marine Protected Areas in The Philippines: Biophysical, Legal, Institutional, and Social Considerations." *Coastal Management* 37: 274–290.
- Lubchenco, J., and K. Grorud-Colvert. 2015. "Making Waves: The Science and Politics of Ocean Protection." *Science (New York, N.Y.)* 350 350: 382–83.
- Magris, R. A., R. L. Pressey, R. Weeks, and N. C. Ban. 2014. "Integrating Connectivity and Climate Change into Marine Conservation Planning." *Biological Conservation* 170: 207–221.
- Malone, D. P., K. Davis, S. I. Lonhart, A. Parsons-Field, J. E. Caselle, and M. H. Carr. 2022. "Large-Scale, Multidecade Monitoring Data from Kelp Forest Ecosystems in California and Oregon (USA)." *Ecology* 103: e3630.
- Marzano, R. J., D. J. Pickering, and J. E. Pollock. 2001. *Classroom Instruction that Works: Research-Based Strategies for Increasing Student Achievement*. Alexandria, VA: Association for Supervision and Curriculum Development.
- McCook, L. J., T. Ayling, M. Cappo, J. H. Choat, R. D. Evans, D. M. De Freitas, M. Heupel, et al. 2010. "Adaptive Management of the Great Barrier Reef: A Globally Significant Demonstration of the Benefits of Networks of Marine Reserves." *Proceedings of the National Academy of Sciences of the United States of America* 107: 18278–85.
- McLeod, E., R. Salm, A. Green, and J. Almany. 2009. "Designing Marine Protected Area Networks to Address the Impacts of Climate Change." *Frontiers in Ecology and the Environment* 7: 362–370.
- McPherson, M. L., D. J. I. Finger, H. F. Houskeeper, T. W. Bell, M. H. Carr, L. Rogers-Bennett, and R. M. Kudela. 2021. "Large-Scale Shift in the Structure of a Kelp Forest Ecosystem Co-Occurs with an Epizootic and Marine Heatwave." *Communications Biology* 4: 1–9.
- Mumby, P. J., I. A. Elliott, C. M. Eakin, W. Skirving, C. B. Paris, H. J. Edwards, S. Enriquez, R. Iglesias-Prieto, L. M. Cherubin, and J. R. Stevens. 2011. "Reserve Design for Uncertain Responses of Coral Reefs to Climate Change." *Ecology Letters* 14: 132–140.
- National Marine Protected Areas Center. 2020. "The MPA Inventory." [https://marineprotectedareas.noaa.gov/dataanalysis/mpainventory/#?~:text=As%20of%20the%202020%20version,Conservation%20of%20Nature%20\(IUCN\)](https://marineprotectedareas.noaa.gov/dataanalysis/mpainventory/#?~:text=As%20of%20the%202020%20version,Conservation%20of%20Nature%20(IUCN).).
- Nickols, K. J., J. W. White, D. Malone, M. H. Carr, R. M. Starr, M. L. Baskett, A. Hastings, and L. W. Botsford. 2019. "Setting Ecological Expectations for Adaptive Management of Marine Protected Areas." *Journal of Applied Ecology* 56: 2376–85.
- Ohayon, S., I. Granot, and J. Belmaker. 2021. "A Meta-Analysis Reveals Edge Effects within Marine Protected Areas." *Nature Ecology & Evolution* 5: 1301–8.
- Ojeda-Martínez, C., J. T. Bayle-Sempere, P. Sanchez-Jerez, F. Salas, B. Stobart, R. Goñi, J. M. Falcón, et al. 2011. "Review of the Effects of Protection in Marine Protected Areas: Current Knowledge and Gaps." *Animal Biodiversity and Conservation* 34: 191–203.
- O'Leary, B. C., M. Winther-Janson, J. M. Bainbridge, J. Aitken, J. P. Hawkins, and C. M. Roberts. 2016. "Effective Coverage Targets for Ocean Protection." *Conservation Letters* 9: 398–404.
- Olsen, E., L. Fonseca, R. Goñi, D. Johnson, S. Katsanevakis, E. Macpherson, P. Weaver, et al. 2013. *Achieving Ecologically Coherent MPA Networks in Europe: Science Needs and Priorities*. European Marine Board.
- Osenberg, C. W., J. S. Shima, S. L. Miller, and A. C. Stier. 2011. "ECOLOGY – Assessing Effects of Marine Protected Areas: Confronting in Space and Possible Solutions." In *Marine Protected Areas: A Multidisciplinary Approach*, edited by J. Claudet, 143–167. Cambridge: Cambridge University Press.
- Ovando, D., J. E. Caselle, C. Costello, O. Deschenes, S. D. Gaines, R. Hilborn, and O. Liu. 2021. "Assessing the Population-Level Conservation Effects of Marine Protected Areas." *Conservation Biology* 35: 1861–70.
- Pinnegar, J. K., N. V. C. Polunin, P. Francour, F. Badalamenti, R. Chemello, M.-L. Harmelin-Vivien, B. Hereu, et al. 2000. "Trophic Cascades in Benthic Marine Ecosystems: Lessons for Fisheries and Protected-Area Management." *Environmental Conservation* 27: 179–200.
- Pondella, D. J., B. E. Gintert, J. R. Cobb, and L. G. Allen. 2005. "Biogeography of the Nearshore Rocky-Reef Fishes at the Southern and Baja California Islands." *Journal of Biogeography* 32: 187–201.
- Pondella, D. J., I. S. Piacenza, J. Claisse, C. Williams, J. Williams, A. Zellmer, and J. Caselle. 2019. "Assessing Drivers of Rocky Reef Fish Biomass Density from the Southern California Bight." *Marine Ecology Progress Series* 628: 125–140.
- Rees, M. J., N. A. Knott, M. L. Hing, M. Hammond, J. Williams, J. Neilson, D. S. Swadling, and A. Jordan. 2021. "Habitat and Humans Predict the Distribution of Juvenile and Adult Snapper (Sparidae: *Chrysophrys auratus*) along Australia's Most Populated Coastline." *Estuarine, Coastal and Shelf Science* 257: 107397.
- Roberts, C. M., B. C. O'Leary, D. J. McCauley, P. M. Cury, C. M. Duarte, J. Lubchenco, D. Pauly, et al. 2017. "Marine Reserves Can Mitigate and Promote Adaptation to Climate Change." *Proceedings of the National Academy of Sciences of the United States of America* 114: 6167–75.
- Rogers-Bennett, L., and C. A. Catton. 2019. "Marine Heat Wave and Multiple Stressors Tip Bull Kelp Forest to Sea Urchin Barrens." *Scientific Reports* 9: 15050.
- Saarman, E., M. Gleason, J. Ugoretz, S. Aíramé, M. Carr, E. Fox, A. Frimodig, T. Mason, and J. Vasques. 2013. "The Role of Science in Supporting Marine Protected Area Network Planning and Design in California." *Ocean & Coastal Management* 74: 45–56.
- Schiel, D. R., and M. S. Foster. 2015. *The Biology and Ecology of Giant Kelp Forests*. Oakland, CA: University of California Press.
- Sciberras, M., S. R. Jenkins, M. J. Kaiser, S. J. Hawkins, and A. S. Pullin. 2013. "Evaluating the Biological Effectiveness of Fully and Partially Protected Marine Areas." *Environmental Evidence* 2(4): 1–31.
- Smith, J. G., C. M. Free, C. Lopazanski, J. Brun, C. R. Anderson, M. H. Carr, J. Claudet, et al. 2023. "A Marine Protected Area

- Network Does Not Confer Community Structure Resilience to a Marine Heatwave across Coastal Ecosystems.” *Global Change Biology* 29: 5634–51. <https://doi.org/10.1111/gcb.16862>.
- Smith, J. G., C. Lopazanski, C. M. Free, J. Brun, C. Anderson, M. H. Carr, J. Claudet, et al. 2025. “Conservation Benefits of a Large Marine Protected Area Network Accrue across Multiple Coastal Ecosystems.” *Conservation Biology* 29 (19): e14435.
- Smith, J. G., D. Malone, and M. H. Carr. 2024. “Consequences of Kelp Forest Ecosystem Shifts and Predictors of Persistence through Multiple Stressors.” *Proceedings of the Royal Society B: Biological Sciences* 291: 20232749.
- Starr, R. M., D. E. Wendt, C. L. Barnes, C. I. Marks, D. Malone, G. Waltz, K. T. Schmidt, et al. 2015. “Variation in Responses of Fishes across Multiple Reserves within a Network of Marine Protected Areas in Temperate Waters.” *PLoS One* 10: e0118502.
- Steneck, R. S., C. B. Paris, S. N. Arnold, M. C. Ablan-Lagman, A. C. Alcala, M. J. Butler, L. J. McCook, G. R. Russ, and P. F. Sale. 2009. “Thinking and Managing outside the Box: Coalescing Connectivity Networks to Build Region-Wide Resilience in Coral Reef Ecosystems.” *Coral Reefs* 28: 367–378.
- Tissot, B. N., W. J. Walsh, and L. E. Hallacher. 2004. “Evaluating Effectiveness of a Marine Protected Area Network in West Hawai’i to Increase Productivity of an Aquarium Fishery.” *Pacific Science* 58: 175–188.
- Van Diggelen, A. D., S. E. Worden, A. J. Frimodig, and S. P. Wertz. 2022. “California’s Lessons Learned and Recommendations for Effective Marine Protected Area Network Management.” *Marine Policy* 137: 104928.
- Vandeperre, F., R. M. Higgins, J. Sánchez-Meca, F. Maynou, R. Goñi, P. Martín-Sosa, A. Pérez-Ruzafa, et al. 2011. “Effects of No-Take Area Size and Age of Marine Protected Areas on Fisheries Yields: A Meta-Analytical Approach.” *Fish and Fisheries* 12: 412–426.
- Watson, D. L., E. S. Harvey, G. A. Kendrick, K. Nardi, and M. J. Anderson. 2007. “Protection from Fishing Alters the Species Composition of Fish Assemblages in a Temperate-Tropical Transition Zone.” *Marine Biology* 152: 1197–1206.
- Wauchope, H. S., T. Amano, J. Geldmann, A. Johnston, B. I. Simmons, W. J. Sutherland, and J. P. G. Jones. 2021. “Evaluating Impact Using Time-Series Data.” *Trends in Ecology & Evolution* 36: 196–205.
- Wernberg, T., K. Krumhansl, K. Filbee-Dexter, and M. F. Pedersen. 2019. “Chapter 3 – Status and Trends for the World’s Kelp Forests.” In *World Seas: An Environmental Evaluation*, 2nd ed., edited by C. Sheppard, 57–78. San Diego, CA: Academic Press.
- White, J. W., L. W. Botsford, A. Hastings, M. L. Baskett, D. M. Kaplan, and L. A. K. Barnett. 2013. “Transient Responses of Fished Populations to Marine Reserve Establishment: Transient Dynamics in Marine Reserves.” *Conservation Letters* 6: 180–191.
- White, J. W., L. W. Botsford, E. A. Moffitt, and D. T. Fischer. 2010. “Decision Analysis for Designing Marine Protected Areas for Multiple Species with Uncertain Fishery Status.” *Ecological Applications* 20: 1523–41.
- Woodley, S., N. Loneragan, and R. Babcock. 2010. *Report on the Scientific Basis for and the Role of Marine Sanctuaries in Marine Planning*. Interdepartmental Committee for the Management of the States Marine Protected Areas.
- Yeager, M., J. W. White, M. H. Carr, D. Malone, and P. T. Raimondi. 2023. “Assessing Connectivity across the California Marine Protected Area Network. Final Report to the California Ocean Protection Council.” 83 pp.
- Zahn, L., J. Claisse, J. Williams, C. Williams, and D. Pondella. 2016. “The Biogeography and Community Structure of Kelp Forest Macroinvertebrates.” *Marine Ecology* 37: 770–785.
- Zellmer, A. J., J. T. Claisse, C. M. Williams, and D. J. Pondella. 2018. “Long-Term, Spatial Marine Harvest Intensity as an Indicator of Human Impact on Shallow Rocky Reef Ecosystems.” *Marine Ecology* 39(S1): e12463.
- Ziegler, S. L., R. O. Brooks, L. F. Bellquist, J. E. Caselle, S. G. Morgan, T. J. Mulligan, B. I. Ruttenberg, et al. 2023. “Collaborative Fisheries Research Reveals Reserve Size and Age Determine Efficacy across a Network of Marine Protected Areas.” *Conservation Letters* 17(2): e13000.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Spiecker, Barbara, Mark H. Carr, Daniel P. Malone, Avrey Parsons-Field, Kathryn Koehn, Daniel J. Pondella II, and Jennifer E. Caselle. 2025. “Measuring Biological Effectiveness across a Very Large, Coherent Network of Coastal Marine Protected Areas.” *Ecological Applications* 35(5): e70074. <https://doi.org/10.1002/eap.70074>